Experiment 7 : Analysis and Visualization of Stock M Data

Aim: Analysis and Visualization of Stock Market Data.

Objective: The dataset contains historical stock price data of a company over a period of time, v attributes such as date, closing price and volume. The goal is to perform time series analysis on stock price data to identify trends, patterns and potential predictors, and to build a model to forec future stock prices.

In [1]:
```
# A52 Dipanshu Ambilkar
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
# Step 1: Import the Stock Prices dataset
df = pd.read_csv("stock.csv")
print("Shape:", df.shape)
df.head()
```

```
Shape: (240, 3)
```

Out[2]:

	date	close	volume
0	2025-01-01	99.68	5928961
1	2025-01-02	101.43	2319311
2	2025-01-03	102.17	7513947
3	2025-01-04	100.67	2603655
4	2025-01-05	96.00	4206855

In [3]:
```
# Step 2: Convert the date column to datetime and set it as index
df["date"] = pd.to_datetime(df["date"])
df = df.set_index("date")
print("Data from", df.index.min(), "to", df.index.max())
df.head()
```

```
Data from 2025-01-01 00:00:00 to 2025-08-28 00:00:00
```

Out[3]:

date	close	volume
2025-01-01	99.68	5928961
2025-01-02	101.43	2319311
2025-01-03	102.17	7513947
2025-01-04	100.67	2603655
2025-01-05	96.00	4206855

In [4]:
```
# Step 3: Line plot of the closing price over time
plt.figure(figsize=(10, 4))
```

```
plt.plot(df.index, df["close"], color="darkblue")
plt.title("Stock Closing Price over Time")
plt.xlabel("Date")
plt.ylabel("Price (Rs)")
plt.grid(True)
plt.show()
```



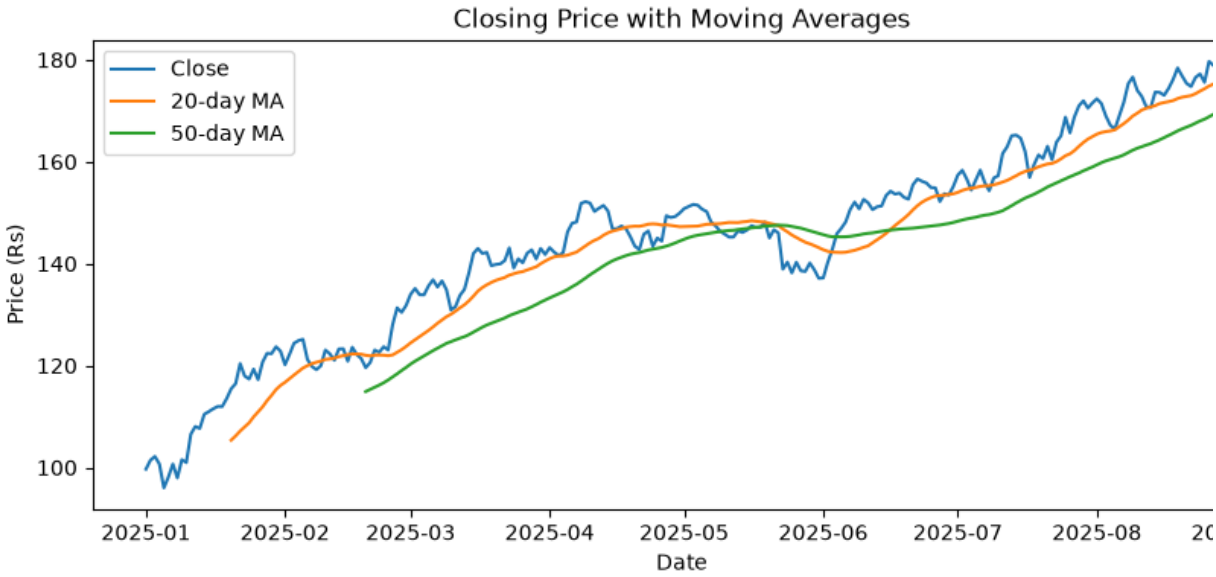
In [5]:
```
# Step 4: Calculate 20-day and 50-day moving averages
df["ma20"] = df["close"].rolling(20).mean()
df["ma50"] = df["close"].rolling(50).mean()
df[["close", "ma20", "ma50"]].tail(10)
```

Out[5]:

date	close	ma20	ma50
2025-08-19	178.40	172.3980	166.0536
2025-08-20	176.83	172.6630	166.4424
2025-08-21	175.37	172.8140	166.7834
2025-08-22	174.76	172.9800	167.1468
2025-08-23	176.55	173.3580	167.5890
2025-08-24	177.23	173.8615	168.0058
2025-08-25	175.59	174.3210	168.3504
2025-08-26	179.68	174.8565	168.8258
2025-08-27	178.90	175.2195	169.3176
2025-08-28	178.95	175.4005	169.7606

In [6]:
```
# Plot closing price with moving averages
plt.figure(figsize=(10, 4))
plt.plot(df.index, df["close"], label="Close")
plt.plot(df.index, df["ma20"], label="20-day MA")
plt.plot(df.index, df["ma50"], label="50-day MA")
plt.title("Closing Price with Moving Averages")
plt.xlabel("Date")
plt.ylabel("Price (Rs)")
```

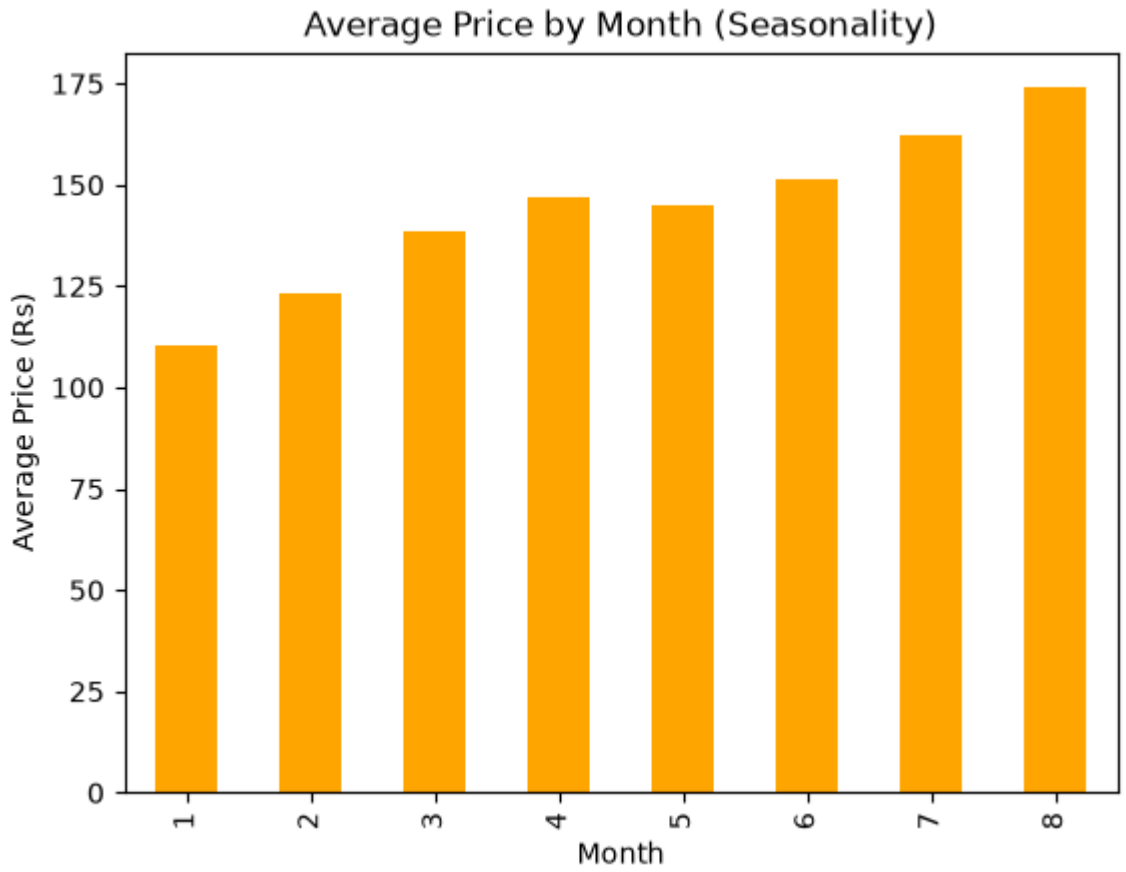
```
plt.legend()
plt.show()
```



In [7]:
```
# Step 5: Seasonality analysis - average closing price by month
df["month"] = df.index.month
monthly = df.groupby("month")["close"].mean()
print("Average closing price by month:")
print(monthly)
```

```
Average closing price by month:
month
1    110.577742
2    123.288571
3    138.346774
4    147.082333
5    145.217097
6    151.381667
7    162.137097
8    173.938929
Name: close, dtype: float64
```

In [8]:
```
# Bar plot of monthly average price
monthly.plot(kind="bar", color="orange")
plt.title("Average Price by Month (Seasonality)")
plt.xlabel("Month")
plt.ylabel("Average Price (Rs)")
plt.show()
```



In [9]:
```
# Step 6: Correlation between stock price and trading volume
print("Price-volume correlation:", round(df["close"].corr(df["volume"]),
plt.scatter(df["volume"], df["close"], color="purple")
plt.title("Volume vs Price")
plt.xlabel("Volume")
plt.ylabel("Price (Rs)")
plt.show()
```

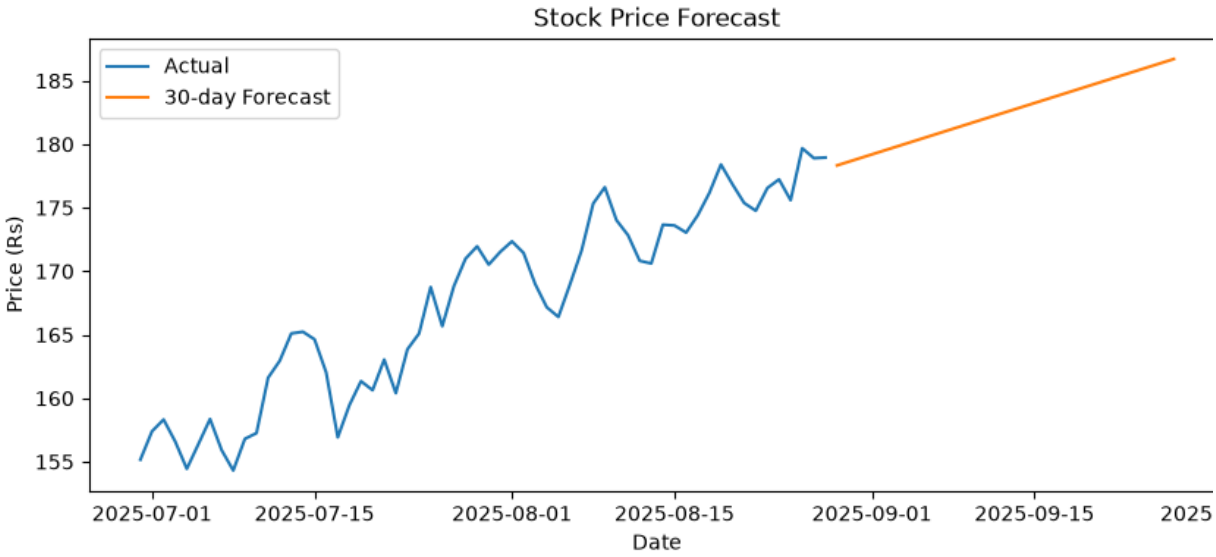
```
Price-volume correlation: -0.038
```



In [10]:
```
# Step 7: Forecast next 30 days using exponential smoothing
smoothed = df["close"].ewm(alpha=0.3, adjust=False).mean()
trend = (smoothed.iloc[-1] - smoothed.iloc[-30]) / 30
future_dates = pd.date_range(df.index[-1] + pd.Timedelta(days=1), periods
forecast = smoothed.iloc[-1] + trend * np.arange(1, 31)
forecast_series = pd.Series(forecast, index=future_dates)
print("Forecast for the next 5 days:")
print(forecast_series.head())
```

```
Forecast for the next 5 days:
2025-08-29    178.330214
2025-08-30    178.618994
2025-08-31    178.907775
2025-09-01    179.196555
2025-09-02    179.485336
Freq: D, dtype: float64
```

In [11]:
```
# Plot actual prices with the 30-day forecast
plt.figure(figsize=(10, 4))
plt.plot(df.index[-60:], df["close"][-60:], label="Actual")
plt.plot(forecast_series.index, forecast_series.values, label="30-day For
plt.title("Stock Price Forecast")
plt.xlabel("Date")
plt.ylabel("Price (Rs)")
plt.legend()
plt.show()
```



Conclusion: Analyzing and visualizing stock market data helps traders and investors track mark performance, identify opportunities, and make data-driven financial decisions. In this experiment moving averages, seasonality and the price-volume relation were analyzed, and future prices we forecast using exponential smoothing.